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LDO_DAO SKY130 Acceptance Test Report

Five-corner ngspice rerun with corrected measurements
and waveform plots

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Executive Summary

The full five-corner SKY130/ngspice acceptance suite was rerun from the updated testbenches. The suite status is **FAIL**. The failing checks are `dc_iq` and `loop_stability_ac`. `dc_iq` still violates the $I_q \leq 3 \text{ uA}$ limit and the low-reference DC output check still fails in several points. `loop_stability_ac` now reports the corrected ngspice phase margin and fails several PVT points near the 40 deg limit.

The earlier apparent large change of I_{q_uA} under static load was a test/reporting error: the old report used total VDD source current as I_{q_uA} , so the 15 uA load current was included. The updated report separates total input current from corrected quiescent current. The earlier PM ≈ 177 deg loop-stability result was also a measurement error: ngspice `vp(...)` returned radians, but the old scalar calculation treated that value as degrees.

Test	Status	CSV metric rows	Failed rows
<code>dc_iq</code>	FAIL	390	75
<code>dynamic_load_tran</code>	PASS	315	0
<code>psrr_ac</code>	PASS	45	0
<code>loop_stability_ac</code>	FAIL	180	14
<code>vout_variation_mc</code>	PASS	260	0

Total CSV rows: 1190. Failed metric rows: 89.

DUT and Conditions

- DUT netlist: `examples/ldo_dao_sky130_from_gf55.sp`.
- Model library: `SKY130_sky130.lib.spice`.
- Corners: `typical -> tt, fff -> ff, ssf -> ss, fsf -> fs, sff -> sf`.
- Output capacitor in all acceptance fixtures: $C_{OUT} = 449 \text{ pF}$.
- Nominal operating point: $V_{DD} = 3.3 \text{ V}$, $V_{REF} = 0.80 \text{ V}$, $I_{BIAS} = 400 \text{ nA}$, $TEMP = 27 \text{ C}$.
- Static load condition: 15 uA.
- Dynamic load condition: 15 uA DC plus 30 mA, 200 ps, 10 MHz pulse load.

Results by Test

DC and Quiescent Current

Status: **FAIL**. Failure reasons: `IQ_HIGH`: 65, `VOUT_LOW`: 10.

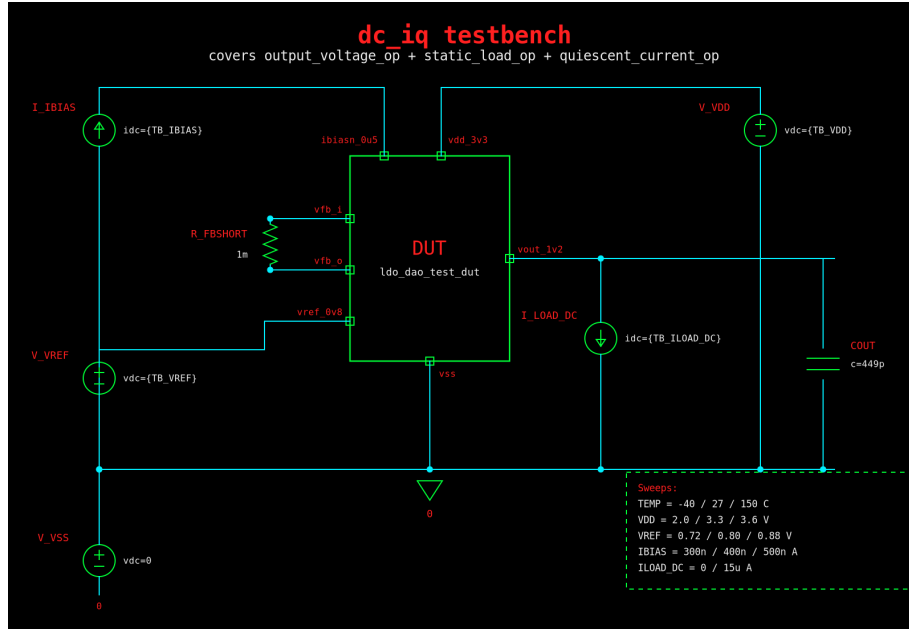


Figure 1: DC regulation and quiescent-current testbench

The DC fixture measures the operating point at `vout_1v2` and the current through the VDD source. ngspice defines current through a voltage source from the positive terminal into the source, so positive DUT consumption is reported as:

$$\begin{aligned} I_{in_total_uA} &= -1e6 * i(V_VDD) \\ I_{q_uA} &= I_{in_total_uA} - 1e6 * I_{LOAD_DC} \end{aligned}$$

For no-load quiescent-current acceptance, `ILOAD_DC = 0`, so $I_{q_uA} = I_{in_total_uA}$. The acceptance checks are:

$$\begin{aligned} 1.08 \text{ V} &\leq V_{out_V} \leq 1.32 \text{ V} \\ I_{q_uA} &\leq 3.0 \text{ uA} \end{aligned}$$

Measured ranges:

Metric	Range
Vout_V	1.07433 .. 1.31663
Iin_total_uA	9.96236 .. 27.1023
Iq_uA	9.93972 .. 12.1207

No-load I_{q_uA} range: 9.962 .. 12.12 uA. Loaded corrected I_{q_uA} range: 9.94 .. 12.1 uA. Loaded total input-current range: 24.94 .. 27.1 uA.

Interpretation: the previous ~10 .. 27 uA quiescent-current range mixed no-load current with total input current under a 15 uA external load. After correction, I_{q_uA}

remains around 10 .. 12 μA ; this is still above the 3 μA limit, but it no longer shows the artificial 15 μA load-current jump.

The DUT netlist contains an internal feedback-divider path of approximately $40.0176 \text{ k}\Omega + 40.0176 \text{ k}\Omega + 40.0176 \text{ k}\Omega$, which alone implies about $1.2 \text{ V} / 120 \text{ k}\Omega \approx 10 \text{ }\mu\text{A}$. That matches the corrected no-load I_{q_uA} and explains why the current limit still fails.

Dynamic-Load Transient

Status: **PASS**. Failure reasons: none.

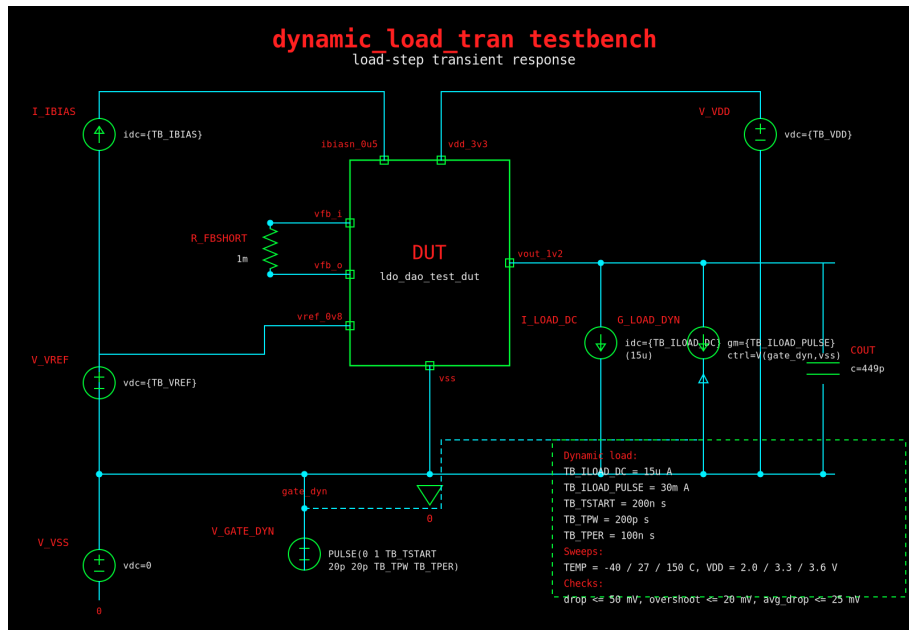


Figure 2: Dynamic-load transient testbench

The transient fixture measures a pre-step level, a minimum, a maximum, and a dynamic average:

```
Vout_pre      = avg(V(vout_1v2), 150 ns .. 190 ns)
Vout_min      = min(V(vout_1v2), 200 ns .. 400 ns)
Vout_max      = max(V(vout_1v2), 200 ns .. 400 ns)
Vout_dyn_avg  = avg(V(vout_1v2), 400 ns .. 900 ns)
```

```
drop_mV       = max(0, 1000 * (Vout_pre - Vout_min))
overshoot_mV  = max(0, 1000 * (Vout_max - Vout_pre))
avg_drop_mV   = max(0, 1000 * (Vout_pre - Vout_dyn_avg))
```

Acceptance limits are $\text{drop_mV} \leq 50$, $\text{overshoot_mV} \leq 20$, and $\text{avg_drop_mV} \leq 25$.

Measured ranges:

Metric	Range
drop_mV	14.699 .. 21.567
overshoot_mV	0 .. 5.977
avg_drop_mV	4.211 .. 8.471
Vout_min_V	1.1702 .. 1.18073

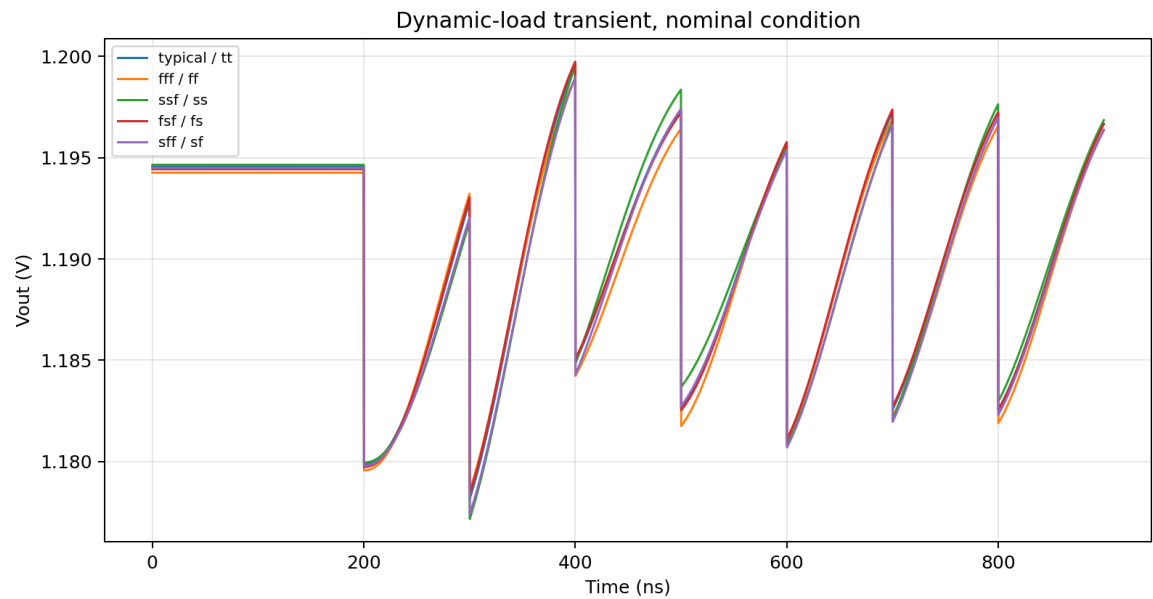


Figure 3: Dynamic-load Vout waveform

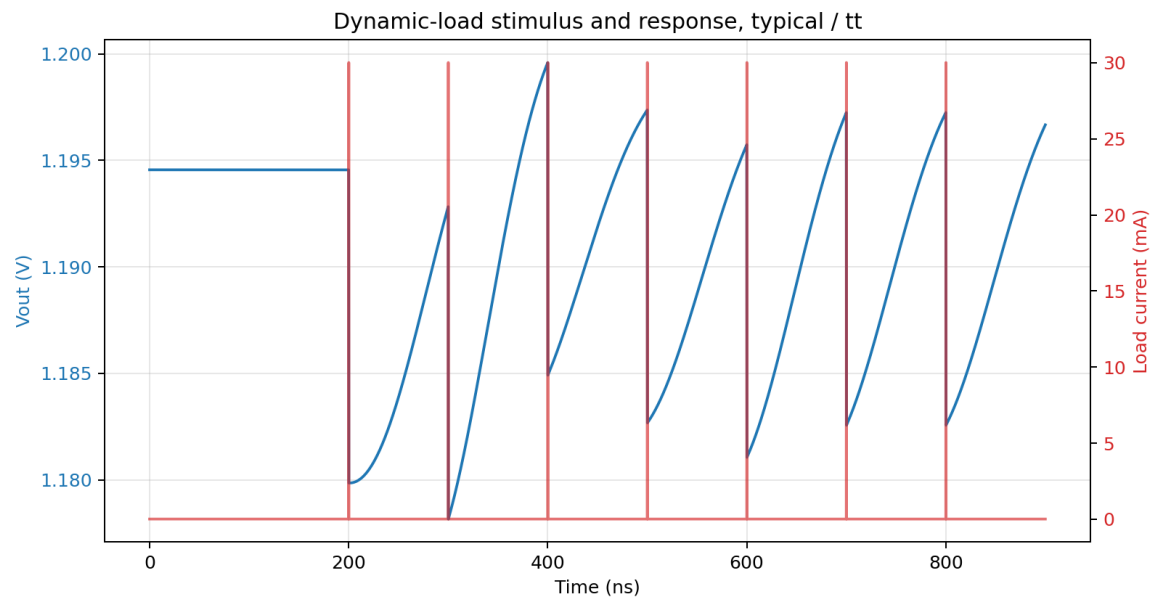


Figure 4: Dynamic-load stimulus and response

PSRR AC

Status: **PASS**. Failure reasons: none.

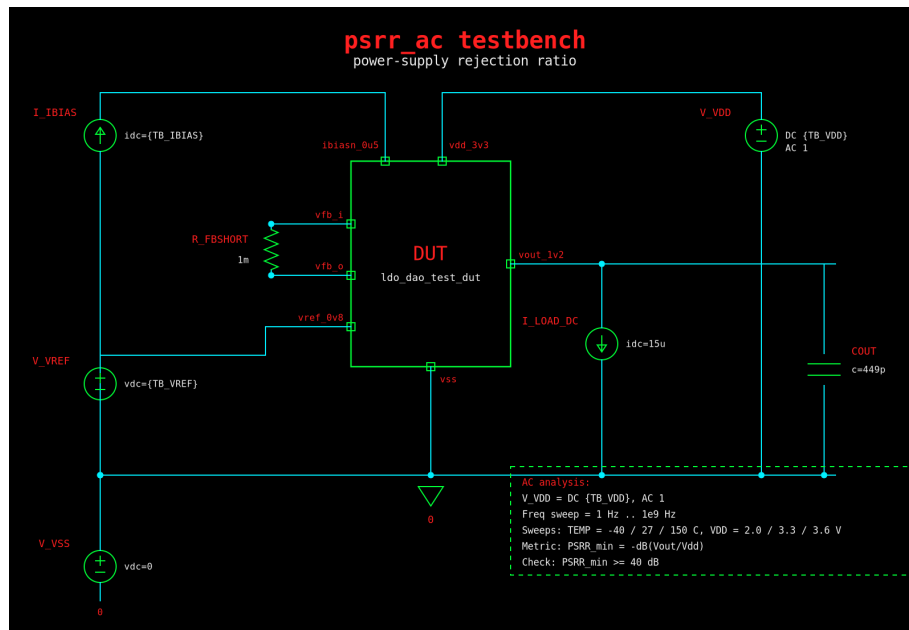


Figure 5: PSRR AC testbench

The PSRR fixture applies AC 1 on V_VDD, so the AC output transfer directly gives Vout/Vdd:

$$\begin{aligned} \text{ratio}(f) &= V(vout_1v2, f) / V(vdd_3v3, f) \\ \text{PSRR_dB}(f) &= -20 * \log_{10}(|\text{ratio}(f)|) \\ \text{PSRR_min_dB} &= \min(\text{PSRR_dB}(f)), f = 1 \text{ Hz} \dots 1 \text{ GHz} \end{aligned}$$

The acceptance limit is $\text{PSRR_min_dB} \geq 40$.

Measured range:

Metric	Range
PSRR_min_dB	41.6887 .. 43.8687

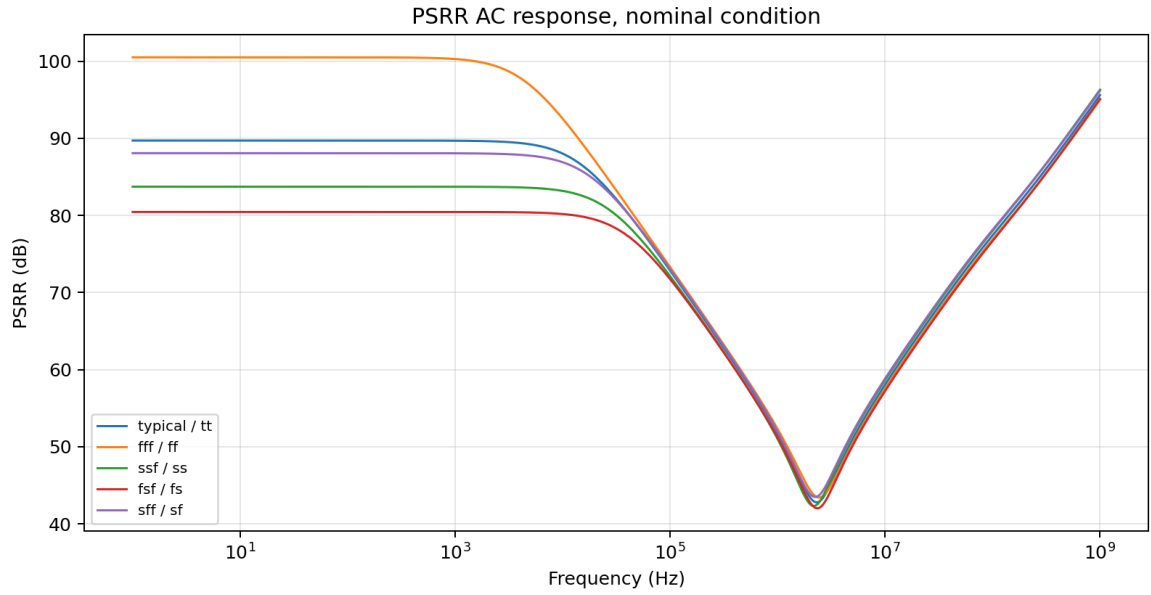


Figure 6: PSRR waveform

Loop Stability AC

Status: **FAIL**. Failure reasons: PM_LOW: 14.

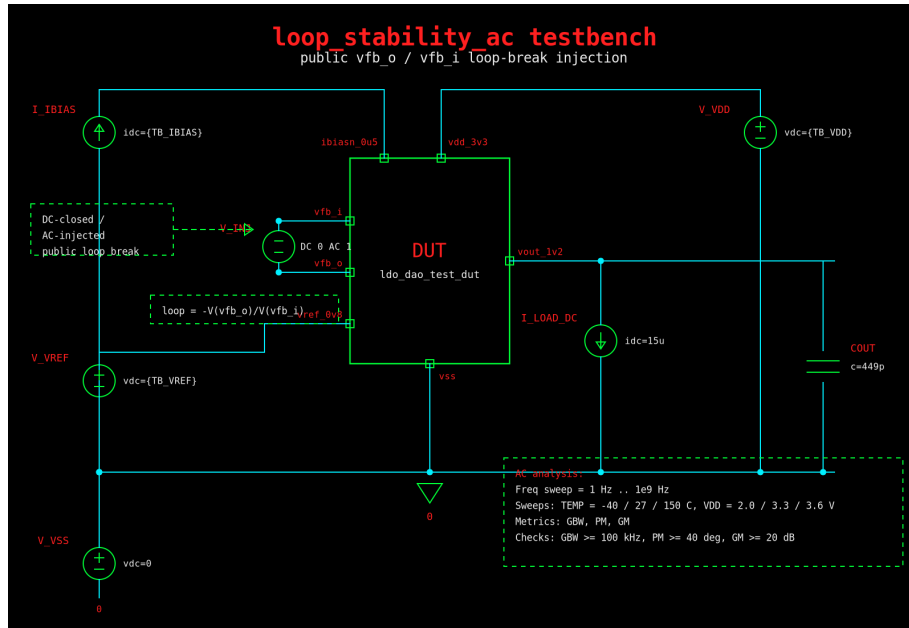


Figure 7: Loop-stability AC testbench

The loop-stability fixture keeps the DC feedback path closed with a zero-volt injection source between the public feedback pins and injects AC 1. The ngspice return-ratio expression is:

$$T(f) = -V(vfb_o, f) / V(vfb_i, f)$$

$$\text{loop_gain_dB}(f) = 20 * \log_{10}(|T(f)|)$$

```
loop_phase_deg(f) = rad_to_deg(wrapped_phase(T(f)))
```

```
GBW_Hz = first f where loop_gain_dB(f) crosses 0 dB falling
```

```
PM_deg = 180 + loop_phase_deg(GBW_Hz)
```

The acceptance limits are $\text{GBW_Hz} \geq 100 \text{ kHz}$, $\text{PM_deg} \geq 40 \text{ deg}$, and $\text{GM_dB} \geq 20 \text{ dB}$.

Measured ranges:

Metric	Range
GBW_Hz	1.4845e+06 .. 2.33599e+06
PM_deg	35.7024 .. 48.9131
GM_dB	23.9618 .. 26.9461

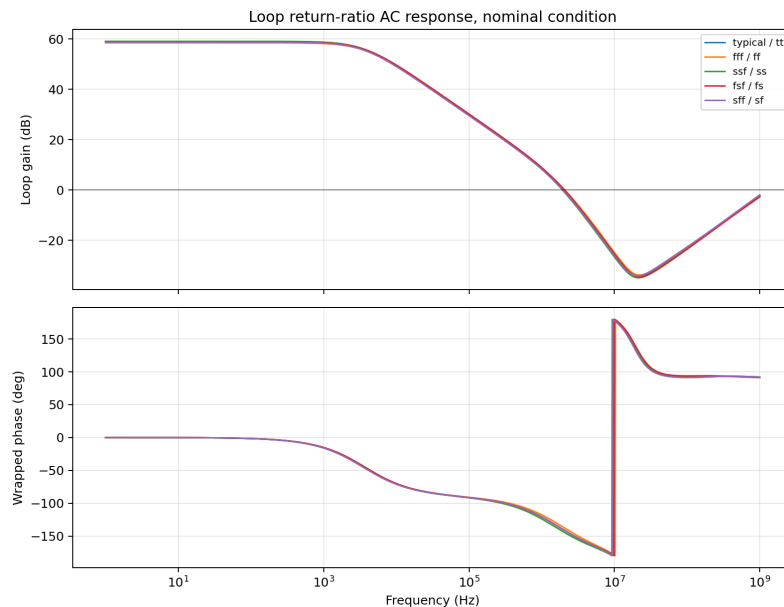


Figure 8: Loop gain and phase waveform

Interpretation: `ngspice vp( . . . )` returns phase in radians in this flow. The measurement control converts it to degrees before calculating phase margin. The loop-stability result should therefore be read from the corrected `PM_deg` range above, not from the legacy report that added a radian phase directly to 180. The failing points are phase-margin failures, while gain bandwidth and gain margin remain above their limits.

Monte Carlo Output Variation

Status: **PASS**. Failure reasons: none.

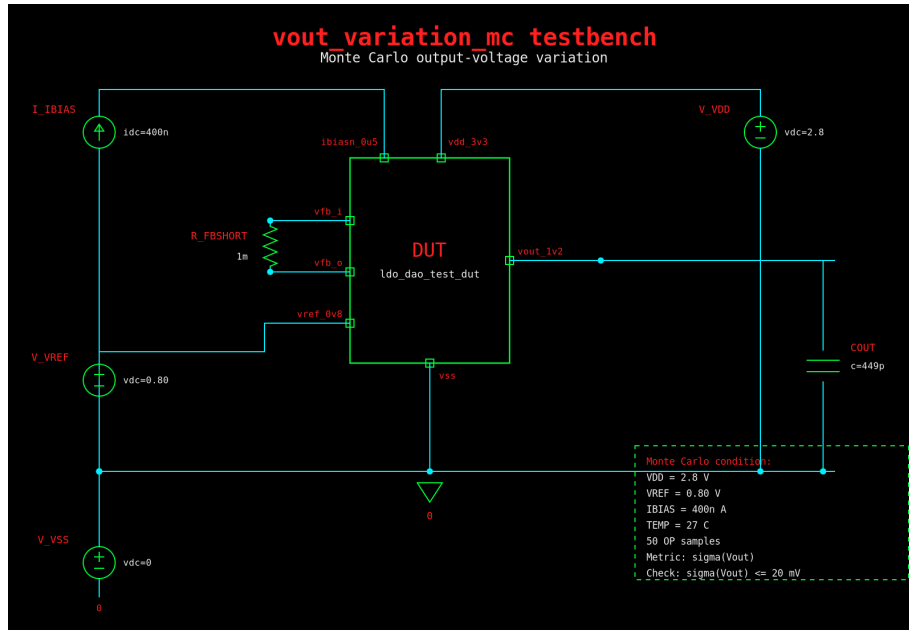


Figure 9: Monte Carlo output-variation testbench

The Monte Carlo fixture runs 50 operating-point samples using the SKY130 mismatch sections and reports:

```
mean_V      = average(Vout_i)
sigma_mV    = 1000 * standard_deviation(Vout_i)
```

The acceptance limit is $\text{sigma_mV} \leq 20$.

Measured ranges:

Metric	Range
mean_V	1.19568 .. 1.19587
sigma_mV	17.3908 .. 17.8922

Conclusions

The test bug behind the apparent load-dependent quiescent-current swing is fixed: total input current and corrected quiescent current are now reported separately.

The DUT still fails DC acceptance because corrected no-load I_{q_uA} is roughly 10 .. 12 μA , above the 3 μA requirement, and low-reference points still violate the output lower bound.

The loop-stability scalar must use degree-converted ngspice phase. A stronger follow-up would still be to compare this public-pin return-ratio setup against a

Middlebrook/Tian-style return-ratio setup or against simulator-native stability analysis.